

Introduction

This project intends to explore how varying attributes of NPC perception can influence individual and crowd behavior. By analyzing movement patterns and group dynamics, we aim to understand how NPCs transition between acting independently and forming collective behaviors in response to environmental stimuli.

Abstract Summary

Main Goal:

- Explore how AI-driven non-playable characters (NPCs) navigate and adapt in crowded environments with a focus on how randomizing their individual attributes (ie. perception cone, environmental preferences) affect crowd behavior and interaction.

How This is Accomplished:

- Through Unreal Engine 5's Environmental Query System (EQS), Behavior Tree, Blueprints, and custom C++ files, we created NPCs with varying behaviors.
- Randomize NPC attributes like perception cone degree and range and other perception features using a normal distribution.
- Tracking NPC movement and log data with time-stamped locations for analysis.
- Structure this data so that external geographic information systems (GIS) can analyze how NPC attributes influence crowd dynamics.

Research Progress

Randomized NPC Perception:

To create NPCs that feel like unique individuals, we decided to give NPCs unique perception areas by varying the degree and range of their EQS cones. This involved allowing each NPC to initiate their EQS with a pair of run-time generated random numbers (one for cone degree and one for cone range). This results in each NPC spawning with different sized EQS cone (see **Figure A**). See Human Variability in NPC Traits for more details on we generated the random numbers.

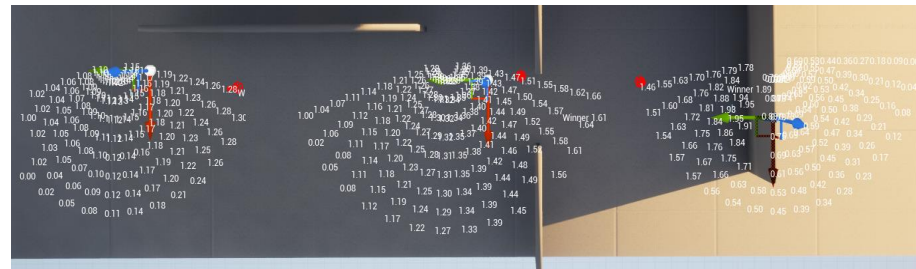


Figure A: different EQS areas for different NPCs

NPC Location Logging:

We developed a system to track and export real-time locations of NPCs in Unreal Engine for future integration with geographic information systems (GIS) software. This involved attaching timed logging logic to NPC actors, allowing us to periodically capture their world position and calculate their distance to a designated goal actor. These distance values were exported to a CSV file over time. By varying different EQS parameters, we analyzed how changes in NPC perception influenced navigational efficiency and crowd behavior. These plots made it easier to observe how changes in EQS parameters impacted NPC movement patterns and goal-reaching behavior.

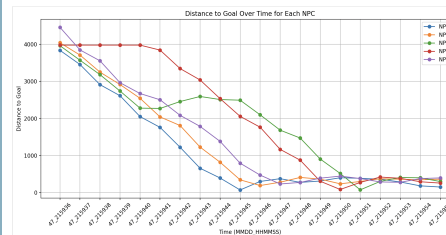


Figure B: NPC Distance to Goal vs Time (cone angle mean = 170°, variance = 25°)

Human Variability in NPC Traits:

To accurately simulate crowds, we decided to vary NPC traits according to a normal distribution (also known as a Bell curve or Gaussian distribution). We used the Marsaglia polar method to convert a pair of Unreal Engine's uniform random variables into a pair of standard normal random variables, then into normal random variables with user specified mean and variance. This was most convenient because we wanted 2 random normal variables (one for cone degree and one for cone range) for each NPC. The process of generating the degree of an NPC's EQS cone at run-time is as seen in **Figure C**.

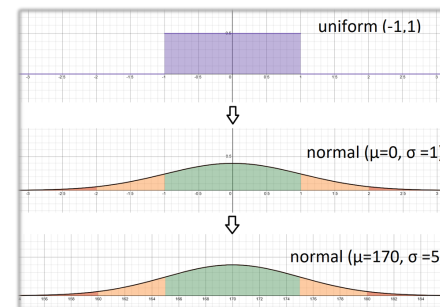


Figure C: conversion from uniform to the desired normal distribution for an NPC's EQS cone degree

Implications

Our research aims to improve NPC simulations to be a more accurate representation of real-life crowds and use the data to analyze how the decision-making process in AI NPC characters can be extended to real world scenarios. Our research can be used in game-development and urban planning applications.

Future Plans

We aim to further develop the uniqueness of NPCs by implementing other randomized sense like smell or hearing. In the future, we also hope to implement the ability to configure EQS specification using an external file outside of Unreal Engine. This will make it easier for the end user to adjust the simulation to their liking without needing the knowledge of Unreal Engine.

References

Barbarash, D. (2025, March 29) *Designing personality – artificial intelligence and digital simulation*. CELA 2025. Portland, Oregon, USA.